RESEARCH PAPER

An Interactive and Reproducible Bioinformatics Platform for the Division of Computational Biology, UCT:

Bridging the Computational Access Divide in African Genomics through Web-Based scRNA-seq and Omics Analysis, with a Roadmap for AI-Augmented Research

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Abstract

Africa bears a disproportionate burden of infectious, genetic, and non-communicable disease, yet remains structurally underrepresented in the computational genomics tools built to address it. South Africa, home to the continent's most productive bioinformatics infrastructure through the Division of Computational Biology (CBIO) at the University of Cape Town and its role as H3ABioNet's primary national node, exemplifies this paradox. CBIO maintains six mature, open-source omics pipelines and serves a pan-African training mandate through the H3Africa consortium — yet every one of these analytical capabilities is locked behind command-line interfaces accessible only to a minority of the researchers who need them most.

This paper presents the design, implementation, and critical evaluation of an Interactive Bioinformatics Platform that directly resolves this access deficit. We describe two production-deployed systems: a guided scRNA-seq Explorer implementing an eight-stage single-cell RNA sequencing pipeline (quality control through automated report generation), and a Variant Analysis Suite providing multi-sample VCF processing with SnpEff functional annotation, ClinVar integration, and ACMG-lite pathogenicity classification. Both are built on CBIO's existing infrastructure, containerised with Docker, and currently accessible at scrna-analysis.streamlit.app and simon-variants.streamlit.app respectively.

Beyond the immediate platform, this paper makes three substantive contributions: first, an empirically grounded analysis of the bioinformatics capacity gap as a structural inequity in South African and African research institutions — not merely a usability inconvenience; second, a rigorous deployment architecture analysis for institutional HPC environments in the context of South African data sovereignty law and the NICIS cyberinfrastructure strategy; and third, a detailed roadmap for the phased integration of artificial intelligence — from foundation model-based cell type annotation to autonomous analysis agents — into the platform over a defined multi-year horizon. The argument advanced throughout is that the computational access barrier in African genomics is an epistemic sovereignty problem: its resolution is a precondition for African researchers to generate, interpret, and own knowledge about African biology.

Keywords: *scRNA-seq; interactive bioinformatics; HPC deployment; bioinformatics capacity; African genomics; H3ABioNet; H3Africa; computational access equity; Streamlit; Scanpy; variant analysis; SnpEff; ClinVar; ACMG; South Africa; NICIS; Docker; Nextflow; AI in genomics; foundation models; reproducible research*

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1. Introduction

1.1 The Problem This Paper Addresses

In 2021, a systematic assessment of bioinformatics capacity across H3ABioNet member institutions documented a finding that was striking not for its novelty but for its persistence: the single most frequently cited barrier to independent genomic data analysis among African researchers was not access to sequencing infrastructure, not shortage of sample material, and not absence of funding — it was the inability to use the analytical software that converts sequencing data into biological knowledge.[1] Across institutions in Nigeria, Kenya, Uganda, Ghana, Ethiopia, and South Africa, researchers who had generated high-quality whole-genome or transcriptomic datasets reported that analysis remained dependent on a small number of bioinformaticians who could operate command-line pipelines — creating queues, delays, and a structural subordination of wet-lab science to computational gatekeeping.

This paper addresses that problem in a specific and tractable form: at the Division of Computational Biology (CBIO) at the University of Cape Town (UCT), which functions as South Africa's primary H3ABioNet node and one of the continent's most capable bioinformatics centres.[2] CBIO maintains six production-grade open-source omics pipelines covering the major modalities of contemporary genomic medicine. Its researchers lead internationally recognised programmes in TB pathogenesis, African population genomics, and viral evolution. And yet, by design, every one of CBIO's analytical capabilities is accessible only to researchers who can operate a Unix command line, navigate an HPC job scheduler, and configure containerised workflows — a profile that excludes the overwhelming majority of the 14 clinical departments in UCT's Faculty of Health Sciences that CBIO nominally serves.

The solution described in this paper is an interactive, reproducible, web-based bioinformatics platform: an interface layer that exposes CBIO's existing computational infrastructure through a guided visual environment accessible to any researcher with a web browser. Two components are already built and deployed in production. This paper documents their design, situates them within the broader structural problem they address, and charts the course — including an explicit AI integration roadmap — toward a platform that can serve not just UCT but the H3ABioNet continental network.

1.2 Why This Is Not a Usability Problem

It would be a mistake to frame the computational access barrier in African genomics as a software usability problem — the kind of thing that better documentation or a friendlier interface could resolve. The problem is structural and epistemic. It concerns who has the capacity to generate knowledge from genomic data, and therefore whose scientific questions get asked, whose populations get studied in depth, and whose

[1] H3ABioNet Consortium (2021). Bioinformatics capacity assessment across African member institutions. H3ABioNet internal report. Available at h3abionet.org.
[2] Mulder N et al. (2016). H3ABioNet, a sustainable pan-African bioinformatics network for human heredity and health in Africa. Genome Research, 26(2), 271–277.

researchers build the expertise to lead the next generation of African biomedical science.

The South African Department of Science and Innovation's White Paper on Science, Technology and Innovation (2022) identifies computational capacity as a strategic national priority precisely because it recognises this dynamic.[3] The H3Africa consortium was founded on the same recognition: that generating African genomic data without building African analytical capacity produces a form of scientific extractivism in which the knowledge value of African biology accrues primarily to institutions in the Global North.[4] The NHREC data governance framework addresses a related dimension: genomic data generated in South African patients must, under the Protection of Personal Information Act (POPIA, 2013) and NHREC guidelines, be subject to South African institutional oversight.[5][6] This creates a positive obligation to build South African computational infrastructure capable of processing that data domestically.

> *The goal is not to make existing bioinformatics tools easier to use. It is to ensure that African researchers have the sovereign capacity to analyse African genomic data, draw African biological conclusions, and build African scientific knowledge — without structural dependence on infrastructure, expertise, or interpretive frameworks developed elsewhere.*

The platform described in this paper is a contribution to that goal. It is not, in the end, a software project. It is a capacity-building intervention with a software implementation.

1.3 Research Objectives and Scope

This paper addresses five interrelated objectives:

1. To document the design, implementation, and evaluation framework of the CBIO Interactive Bioinformatics Platform, with particular focus on the scRNA-seq Explorer and Variant Analysis Suite as the first two production-deployed components.
2. To provide an empirically grounded analysis of the bioinformatics capacity gap in South African and African research institutions, with specific reference to H3ABioNet capacity assessments, DSI/NRF strategy documents, and the peer-reviewed literature on African computational biology.
3. To conduct a rigorous deployment and hosting architecture analysis for the South African institutional context, incorporating POPIA data sovereignty requirements, NICIS cyberinfrastructure strategy, and the specific characteristics of CBIO's iLifu HPC environment.

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[3] Department of Science and Innovation (DSI), South Africa (2022). White Paper on Science, Technology and Innovation. Pretoria: DSI. Available at dst.gov.za.
[4] Mulder N et al. (2018). H3Africa: Current Perspectives in Enabling African Genomics. Emerging Topics in Life Sciences, 2(4), 629–636.
[5] NHREC (2015). Ethics in Health Research: Principles, Processes and Structures (2nd ed.). Department of Health, South Africa. Chapter 9: Governance of Health Research Data.
[6] Protection of Personal Information Act (POPIA) No. 4 of 2013. Government Gazette of South Africa, 2013. Available at justice.gov.za.

4. To propose and justify a phased AI integration roadmap — from immediate applications in automated cell type annotation and variant interpretation to longer-term autonomous analysis and multi-modal biological reasoning — grounded in the current state of foundation models in genomics.
5. To establish an evaluation framework by which the platform's impact on researcher independence, reproducibility, and CBIO's grant competitiveness can be measured and reported.

1.4 Paper Structure

Section 2 analyses the bioinformatics capacity gap in South Africa and Africa, grounding the platform's rationale in evidence. Section 3 reviews the literature on interactive platforms, African genomics infrastructure, and AI in biological analysis. Section 4 establishes CBIO's institutional context. Section 5 presents the system architecture. Sections 6 and 7 describe the scRNA-seq Explorer and Variant Analysis Suite in depth. Section 8 provides the deployment and data sovereignty analysis. Section 9 presents the AI integration roadmap. Section 10 addresses the evaluation framework. Section 11 details the implementation roadmap. Section 12 covers resources and funding pathways. Section 13 offers conclusions.

2. The Bioinformatics Capacity Gap in South Africa and Africa: An Evidence-Based Analysis

2.1 Quantifying the Deficit: What the Data Shows

The bioinformatics capacity gap in Africa is not a rhetorical claim — it is a documented structural condition measurable across multiple dimensions. The most comprehensive recent analysis, the H3ABioNet capacity assessment (2021), surveyed 45 institutions across 21 African countries.[7] Its findings warrant direct engagement: fewer than 30% of surveyed institutions had more than two resident bioinformaticians. Fewer than 20% had a dedicated computational server accessible to non-bioinformatician researchers. In 67% of institutions, researchers reported that analysis turnaround times of more than two weeks for standard omics datasets were normal. These are not descriptions of inconvenience; they are quantifications of research delay at a scale that materially affects Africa's ability to respond to infectious disease outbreaks, advance genomic medicine, and produce competitive grant applications.

South Africa sits at the productive end of this distribution but is not immune to its dynamics. UCT's Faculty of Health Sciences — 14 clinical departments, several hundred active researchers and postgraduate students — is served by a small number of dedicated bioinformaticians shared across IDM and CIDRI. The ratio of researchers to bioinformaticians creates the same structural bottleneck observed continent-wide, albeit at a higher absolute throughput. A 2022 study on barriers to bioinformatics education across African institutions identified four recurring structural constraints: insufficient local expertise, absence of user-friendly tools adapted to African research contexts, unreliable high-speed connectivity, and lack of domain-specific training materials.[8] Three of these four are directly addressable through the platform described in this paper.

2.2 South Africa's National Cyberinfrastructure: Capability and Access Gap

South Africa's investment in research cyberinfrastructure is, by African standards, substantial. The National Integrated Cyberinfrastructure System (NICIS), managed by the CSIR, provides the Centre for High Performance Computing (CHPC) in Cape Town — a facility with approximately 29,000 CPU cores and dedicated storage for South African research institutions.[9] UCT's iLifu HPC cluster, managed locally, provides

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[8]Abebe T et al. (2022). Barriers to bioinformatics education and training in low-resource African countries: a mixed-methods survey. PLOS Computational Biology, 18(9), e1010444.
[9]NICIS/CHPC (2023). National Integrated Cyberinfrastructure System Annual Report 2022–2023. Council for Scientific and Industrial Research, Pretoria.

additional dedicated compute for the faculty. The South African National Research Network (SANReN) provides high-bandwidth connectivity between campuses.[10]

These are real assets. The gap is not in physical infrastructure but in the interface between that infrastructure and its potential user base. NICIS/CHPC compute is allocated through a formal project application process; iLifu access requires institutional account provisioning, SSH client configuration, and job scheduler familiarity. These requirements create a de facto access filter that restricts computational research to a technically proficient minority — precisely the filter that the platform described in this paper is designed to remove.

The DSI White Paper on Science, Technology and Innovation (2022) explicitly identifies this interface problem as a national strategic priority, calling for investments that make high-performance computing accessible to “broad research communities including early-career researchers and those in non-computational disciplines.”[11] The NICIS strategy for 2023–2028 similarly identifies user-facing platforms and web-based access as development priorities.[12] The platform described in this paper is not peripheral to this national agenda; it is an implementation of it.

2.3 The scRNA-seq Gap as a Representative Case

Among the methodological gaps in CBIO's current infrastructure, the absence of single-cell RNA sequencing analysis capability is particularly instructive. scRNA-seq has undergone a transformation from specialist technique to mainstream method in the period 2017–2025: the number of cells profiled in published single-cell studies grew from approximately 10,000 in 2017 to over 70 million by 2023, and the method is now standard in TB immunology, HIV pathogenesis, malaria research, and cancer biology — all research areas central to IDM and CIDRI's programmes.[13]

The absence of a usable scRNA-seq analysis interface at CBIO means that researchers at IDM and CIDRI who generate scRNA-seq data — using the considerable sequencing infrastructure available at UCT — currently face one of three options: learn to operate Scanpy or Seurat through the command line (a months-long investment for a wet-lab researcher); commission analysis from CBIO bioinformaticians (creating queue and bottleneck); or export the data to collaborators at institutions abroad where accessible tools exist (creating data sovereignty and analytical ownership problems). None of these is acceptable as a permanent solution. The scRNA-seq Explorer described in this paper provides the fourth option: an accessible, guided analysis environment that keeps data, analysis, and interpretive ownership at CBIO.

2.4 Reproducibility and the Knowledge Deficit

A dimension of the capacity gap that receives insufficient attention in the bioinformatics literature is its impact on reproducibility and therefore on research

[10]NRF (2023). South African Research Infrastructure Roadmap (SARIR). National Research Foundation, Pretoria. Available at nrf.ac.za.

[13]Luecken MD & Theis FJ (2019). Current best practices in single-cell RNA-seq analysis: a tutorial. Molecular Systems Biology, 15(6), e8746.

credibility. The 2016 Baker survey in Nature — in which more than 1,500 scientists reported that over 70% of their colleagues' experiments could not be reproduced — established reproducibility as a systemic problem in biomedical science.[14] In African research institutions, the problem is compounded by the informal analytical environments that the capacity gap produces: analyses conducted ad hoc outside validated pipelines, without parameter records, using software versions that cannot be tracked, and producing outputs that cannot be regenerated.

This is not merely a methodological inconvenience. Research funders — NRF, H3Africa, NIH, Wellcome — increasingly require demonstrable reproducibility as a condition of grant awards. Publications in high-impact journals require data and code availability. The inability to demonstrate reproducible analysis is a direct competitive disadvantage for African researchers in global research funding competitions. The platform addresses this through mandatory provenance recording: every analysis generates a parameter audit log, a JSON configuration export, and a structured methods report — making reproducibility not an optional virtue but an automatic by-product of using the platform.

[14]Baker M (2016). 1,500 scientists lift the lid on reproducibility. Nature, 533(7604), 452–454.

3. Literature Review: Platforms, African Genomics Infrastructure, and AI in Biological Analysis

3.1 Interactive Bioinformatics Platforms: What Exists and Why It Is Not Sufficient

The development of web-based bioinformatics platforms has a substantial history anchored in the Galaxy project, which established in 2010 that accessible interfaces can preserve analytical rigour without sacrificing methodological depth.[15] Galaxy now supports millions of analyses annually across thousands of institutions worldwide and has been deployed at multiple African sites through H3ABioNet. Its limitations in the CBIO context are instructive rather than critical: Galaxy is a general workflow execution platform; it does not carry CBIO's specific pipeline implementations, does not incorporate African variant frequency databases, and does not provide the TB-specific and pathogen-focused analytical modules that define CBIO's research identity. Using Galaxy at CBIO would mean rebuilding CBIO's pipelines in Galaxy's workflow language — a substantial duplication of already-excellent existing work.

The Broad Institute's Terra platform addresses large-scale variant analysis against cloud compute, with deep GATK integration. Its dependency on Google Cloud infrastructure creates both cost and data sovereignty complications for South African academic use. The CELLxGENE platform[16] provides interactive single-cell exploration but targets pre-processed, already-annotated datasets — it is a visualisation tool for completed analyses, not a guided analysis environment for researchers beginning with raw count matrices. The 10x Genomics Loupe Cell Browser is proprietary and platform-locked to Chromium-generated data. None of these tools addresses the requirement — specific to CBIO's context — of a guided, end-to-end analysis environment that integrates with CBIO's existing pipelines, runs on institutional HPC infrastructure, and is accessible to researchers with no prior bioinformatics experience.

The closest conceptual precedent for the approach taken in this paper is the development of domain-specific platform extensions at well-resourced institutions: the Wellcome Sanger Institute's web-based variant interpretation portal, EMBL-EBI's Expression Atlas interactive explorer, and the Broad Institute's Single Cell Portal. These demonstrate that institution-specific, deep-integration analysis platforms can coexist with generic tools and serve distinct user communities. The contribution of this paper is to demonstrate that the same approach is viable — and necessary — in an African institutional context operating under different resource, connectivity, and data sovereignty constraints.

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[15]Goecks J, Nekrutenko A & Taylor J (2010). Galaxy: a comprehensive approach for supporting accessible, reproducible, and transparent computational research. Genome Biology, 11(8), R86.
[16]Chan Zuckerberg Initiative (2021). CELLxGENE: a tool that lets you explore single-cell data. Chan Zuckerberg Initiative Blog. Available at chanzuckerberg.com.

3.2 African Bioinformatics: The Literature on Capacity and Infrastructure

The literature on bioinformatics capacity in Africa has grown substantially over the past decade, reflecting the expansion of the H3Africa consortium and its associated training initiatives. Mulder et al. (2016) documented the establishment of H3ABioNet as a sustainable pan-African bioinformatics network[17] — a paper that both established the governance model for the network and acknowledged the capacity asymmetries it was designed to address. Baichoo et al. (2018) provided a detailed analysis of the infrastructure and training challenges facing H3ABioNet member institutions.[18] Siwo et al. (2021) argued in Nature Methods that building a bioinformatics foundation in Africa requires not just training more bioinformaticians but creating accessible tools that extend the reach of the bioinformaticians who exist.[19]

Abebe et al. (2022) conducted a mixed-methods survey specifically on barriers to bioinformatics education in low-resource African settings, identifying tool inaccessibility and absence of locally adapted training materials as the two most consequential constraints.[20] The African Genomic Medicine Training Initiative (2022) documented the specific challenge of translating genomic research capacity into clinical genomic medicine practice in African health systems, noting that the absence of clinician-accessible variant interpretation tools is a major barrier to implementation.[21]

These papers establish the empirical foundation for the platform described in this paper. The gap is documented; the need is demonstrated; what has been missing is a production-deployed solution that operates within the specific institutional and infrastructure constraints of a leading African computational biology centre. This paper provides that solution.

3.3 scRNA-seq Methods: State of the Field

The single-cell RNA sequencing analytical ecosystem has reached a degree of methodological maturity that makes standardised best-practice pipelines possible. Luecken and Theis (2019) established the canonical best-practices workflow for single-cell analysis: ambient RNA correction, per-cell QC filtering, normalisation, highly variable gene selection, dimensionality reduction (PCA), neighbourhood graph construction, community detection, UMAP visualisation, and cell type annotation.[22]

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[18]Baichoo S et al. (2018). Developing Bioinformatics Capacity in Africa: Building a Sustainable Pan-African Bioinformatics Network (H3ABioNet). PLOS Computational Biology, 14(8), e1006266.
[19]Siwo GH et al. (2021). Building a Foundation for Bioinformatics in Africa. Nature Methods, 18(5), 459–461.

[21]African Genomic Medicine Training Initiative (AGMT, 2022). Genomic medicine implementation in African health systems: challenges and opportunities. Lancet Global Health, 10(4), e467–e469.

This workflow, implemented in Scanpy[23] and Seurat[24], has become the reference standard for single-cell transcriptomic analysis across biological systems.

The Leiden community detection algorithm (Traag et al., 2019) represents the current methodological standard for cell clustering, offering mathematical guarantees on cluster quality that the earlier Louvain algorithm did not provide.[25] Doublet detection methods (DoubletFinder, Scrublet) and ambient RNA correction (SoupX, CellBender) have matured to production-grade reliability and are incorporated as recommended preprocessing steps.

The most rapidly evolving area of scRNA-seq methodology is automated cell type annotation — the step at which clusters identified by graph-based community detection are assigned biological identities. Current approaches range from manual annotation using marker gene databases (CellMarker, PanglaoDB), through reference atlas-based label transfer using tools such as Seurat's TransferData or scANVI[26], to fully automated deep learning-based annotation using foundation models trained on large single-cell atlases.[27] This progression directly informs the AI integration roadmap presented in Section 9 of this paper.

3.4 Variant Analysis: Standards and the African Context

The ACMG/AMP variant classification guidelines (Richards et al., 2015) provide the international standard for pathogenicity classification in clinical genomics.[28] Their implementation in automated tools, however, carries a systematic bias: the population frequency databases that inform criteria such as PM2 (absent or extremely rare in population databases) and BA1 (allele frequency >5%) are predominantly derived from European and East Asian populations. gnomAD v2.1.1, the most widely used population frequency resource, derives 64% of its alleles from individuals of European ancestry.[29] For African populations — which harbour substantially higher genetic diversity and therefore more private variants and lower allele frequencies across the genome — applying gnomAD-derived thresholds systematically overpredicts pathogenicity for benign population-specific variants.

This is not a marginal concern. It is a fundamental limitation of current variant interpretation practice as applied to African patients and research participants. The Variant Analysis Suite described in this paper addresses this limitation through explicit annotation of gnomAD African subpopulation frequencies, flagging variants where the overall gnomAD frequency is low but the African-specific frequency is substantially

[23] Wolf FA, Angerer P & Theis FJ (2018). SCANPY: large-scale single-cell gene expression data analysis. Genome Biology, 19(1), 15.
[24] Hao Y et al. (2021). Integrated analysis of multimodal single-cell data. Cell, 184(13), 3573–3587.
[25] Traag VA, Waltman L & Van Eck NJ (2019). From Louvain to Leiden: guaranteeing well-connected communities. Scientific Reports, 9(1), 5233.
[26] Lotfollahi M et al. (2023). Mapping single-cell data to reference atlases by transfer learning. Nature Biotechnology, 40, 121–130. [scANVI / scVI transfer learning]
[27] Eraslan G et al. (2022). Single-nucleus cross-tissue molecular reference maps toward understanding disease gene function. Science, 376(6594), eabl4290.
[28] Richards S et al. (2015). Standards and guidelines for the interpretation of sequence variants. Genetics in Medicine, 17(5), 405–424.
[29] Karczewski KJ et al. (2020). The mutational constraint spectrum quantified from variation in 141,456 humans. Nature, 581(7809), 434–443. [gnomAD v2.1.1]

higher — a pattern indicative of population-specific benign variation rather than pathogenicity. This African-contextualised variant interpretation is one of the platform's most distinctive research contributions.

3.5 AI in Genomics: From Tools to Foundation Models

Artificial intelligence has progressed in biological applications from narrow predictive models to large-scale foundation models capable of multi-task biological reasoning.[30] The trajectory of this progression is directly relevant to the platform's AI roadmap. AlphaFold2 (Jumper et al., 2021)[31] demonstrated that a neural network trained on evolutionary co-variation patterns could predict protein structure with near-experimental accuracy — a result that validated the biological information content available to large-scale deep learning. In the single-cell domain, scVI, scANVI, and related deep generative models[32] demonstrated that neural network embeddings of gene expression data capture biologically meaningful representations transferable across datasets, donors, and experimental conditions.

The most significant recent development for the platform's AI roadmap is the emergence of single-cell foundation models trained on hundreds of millions of cells from diverse tissues and organisms: scGPT (Cui et al., 2024), Geneformer (Theodoris et al., 2023), and CellPLM represent a new class of model that can perform cell type annotation, gene regulatory inference, and perturbation response prediction without dataset-specific fine-tuning. These models represent the near-term basis for automated cell type annotation in the platform's AI integration roadmap (Section 9). Bommasani et al. (2022)[33] provide the foundational conceptual framework for understanding foundation models — models trained at scale on diverse data that can be adapted to many downstream tasks — as the appropriate paradigm for AI integration in the biological sciences.

[30] Rajpurkar P, Chen E, Banerjee O & Topol EJ (2022). AI in health and medicine. Nature Medicine, 28(1), 31–38.
[31] Jumper J et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583–589.

[33] Bommasani R et al. (2022). On the Opportunities and Risks of Foundation Models. arXiv:2108.07258 [cs.LG]. Centre for Research on Foundation Models, Stanford University.

4. CBIO: Institutional Context and Research Infrastructure

4.1 Research Programmes and Their Computational Requirements

CBIO's research identity is defined by two primary programmes whose computational requirements map directly onto the platform components described in this paper. Professor Nicola Mulder's programme spans TB pathogenesis via comparative genomics and host-pathogen biological networks; human genetic factors in disease susceptibility; African population genetic diversity; and microbiome effects on immunity and disease. Each of these research threads generates a distinct computational requirement: comparative genomics requires efficient multi-genome alignment and variant identification across divergent bacterial strains; host-pathogen biology requires single-cell resolution of immune cell states during infection; population genetics requires African-contextualised variant frequency analysis; and microbiome research requires 16S rRNA and metagenomic diversity analysis.

Associate Professor Darren Martin's programme in viral evolution — built around the Recombination Detection Program (RDP), a globally used tool for identifying genetic recombination in viral genomes — requires phylogenetic visualisation, recombination event mapping, and phylogeographic analysis tools. RDP's widespread use in the global viral genomics community makes CBIO a natural site for the development of interactive visualisation interfaces that can extend RDP's reach to researchers who generate recombination analysis outputs but lack the skills to visualise them programmatically.

4.2 The Existing Pipeline Infrastructure

Pipeline	Data Type	Primary Research Beneficiary	Platform Enhancement
Metagenomics (WGS)	Whole-genome shotgun	Prof. Mulder: microbiome & immunity	Interactive species composition, diversity metrics, QC dashboards
16S rRNA Amplicon	Microbiome profiling	Prof. Mulder: microbiome studies	Visual OTU/ASV explorer, alpha/beta diversity, automated reports
RNAseq (Human)	Bulk transcriptomics	IDM/CIDRI: expression studies	Guided DE analysis, pathway enrichment, interactive volcano plots
Proteomics	Protein expression	IDM/CIDRI: protein abundance	Volcano plots, GO enrichment, interactive abundance dashboards

Pipeline	Data Type	Primary Research Beneficiary	Platform Enhancement
Array Genotyping	SNP arrays / GWAS	Prof. Mulder: population genetics	PCA, population stratification, GWAS visualisation, LD analysis
Human WGS & Exome	Variant discovery	Prof. Mulder / IDM/CIDRI clinical	Variant browser, African-contextualised ClinVar/ACMG annotation

Table 1. CBIO's six existing open-source omics pipelines and their alignment with platform components and research programmes.

4.3 H3ABioNet: The Continental Mandate

CBIO's role as South Africa's primary H3ABioNet node carries obligations that extend well beyond UCT's campus. H3ABioNet is constituted as the bioinformatics training and support infrastructure for the H3Africa consortium[34] — a network that spans more than 30 research institutions across sub-Saharan Africa and generates the largest coordinated collection of African genomic data in existence.[35] CBIO's training workshops bring researchers from across the continent to UCT; its pipeline documentation and training materials are used at institutions in Nigeria, Kenya, Uganda, Ghana, and beyond.

The training mandate creates a specific and demanding requirement for any platform deployed at CBIO: it must be accessible to researchers returning to institutions that may not have UCT-level computational infrastructure, stable high-speed connectivity, or resident bioinformatics support. This rules out deployment models that require institutional VPN access, high-bandwidth data transfer, or complex local software installation. The platform's architecture — web-accessible, browser-based, with data processing occurring server-side — satisfies these constraints by design. A researcher who attends a CBIO workshop and uses the platform during training retains access to the same tool from their home institution via any internet connection.

4.4 Collaborative Ecosystem: IDM, CIDRI-Africa, and Beyond

CBIO operates within an exceptionally rich collaborative ecosystem. IDM and CIDRI-Africa together represent one of Africa's largest concentrations of infectious disease research capacity, with research programmes in TB (including the largest TB cohort study in the world, the RePORT consortium), HIV, malaria, helminth infections, and emerging pathogens. Both institutes generate large-scale omics data — transcriptomics, genomics, epigenomics — that requires CBIO-quality analysis. The dedicated bioinformaticians embedded within IDM and CIDRI serve as advanced users of CBIO's platform: capable of configuring complex analyses, but dependent on CBIO for the underlying pipeline infrastructure.

This collaborative structure defines the platform's primary user tiers: IDM/CIDRI bioinformaticians as power users who configure and validate analyses; IDM/CIDRI wet-lab researchers and clinicians as guided users who run pre-configured workflows and interpret outputs; and H3ABioNet training participants as learners who use the platform to develop analytical independence. Designing for all three tiers simultaneously requires a layered interface architecture — simple defaults that work out of the box, with parameter access and configuration depth available to those who need it.

5. System Architecture and Design Philosophy

5.1 The Interface-Infrastructure Separation Principle

The platform's foundational architectural decision is the strict separation of the interface layer from the computation layer. This separation — implemented through an abstract execution backend that can target local processes, SLURM job submission, or cloud compute APIs without interface code modification — is not merely an engineering convenience. It is the mechanism by which the platform achieves its primary design goal: making CBIO's existing pipelines accessible without replacing or replicating them.

This principle distinguishes the platform from tools that attempt to implement analysis from scratch in a web framework — an approach that creates parallel, unmaintained implementations of algorithms that CBIO's existing pipelines already implement correctly and efficiently. Every analytical step in the platform's scRNA-seq Explorer and Variant Analysis Suite invokes the same algorithmic code that would be executed from the command line: the interface layer handles user input, parameter validation, progress reporting, and output presentation. The computation layer is unchanged. This ensures that a result produced through the web interface is identical to the result that a command-line expert would produce with the same parameters — a property essential for reproducibility and for user trust.

5.2 Technology Stack

Layer	Technology	Justification
Frontend / UI	Streamlit (Python)	Rapid development of data-science UIs with native Python scientific library integration; session state management for multi-step guided workflows; growing adoption in academic bioinformatics
Data Processing	Scanpy, Pandas, NumPy, SciPy	scverse ecosystem alignment; AnnData sparse matrix representation for memory efficiency; seamless integration with Python ML/AI frameworks
Visualisation	Plotly (interactive), Matplotlib/Seaborn (publication)	Interactive zoom/pan/hover for exploratory analysis; publication-quality static export (PNG, SVG, PDF) for manuscripts and grants
Variant Annotation	SnpEff, ClinVar API, scikit-allel	Industry-standard consequence prediction; real-time clinical significance retrieval; African population frequency analysis
Report Generation	ReportLab (PDF), Jinja2 (HTML)	Automated structured reports with full parameter provenance; clinical-grade formatting; exportable JSON configuration for reproducibility

Layer	Technology	Justification
Containerisation	Docker, Singularity	Reproducible execution environments; Singularity for HPC root-free operation; Docker Hub publication for open-source distribution
Workflow Management	Nextflow	Native SLURM scheduler integration; cloud executor support; aligned with existing CBIO pipeline infrastructure (Di Tommaso et al., 2017)
AI Integration (Phase 2+)	scVI, scANVI, foundation models (scGPT)	Transfer learning for cell type annotation; foundation model inference for biological interpretation (see Section 9)

Table 2. Platform technology stack with selection justification. AI integration components are annotated to Section 9.

5.3 Three-Tier Architecture

The platform implements a three-tier architecture: presentation (Streamlit web application), business logic (Python analysis modules with abstract execution backend), and computation (HPC/cloud execution engines). The presentation tier handles all user-facing concerns: data upload with format validation, parameter configuration with evidence-based defaults, real-time analysis progress via SLURM job status polling, results visualisation, and report download. The business logic tier orchestrates the analytical workflow, manages state persistence across session resumption, validates parameter combinations for biological consistency, and handles error recovery with user-legible diagnostics. The computation tier executes analytical algorithms — either in-process for small datasets in development, or via SLURM job submission for production HPC workloads.

The tiers communicate through a documented internal API that abstracts execution environment differences. The same Python analysis module can be invoked directly (local), submitted as a SLURM batch script (HPC), or executed in a cloud container (cloud PaaS) by changing a single configuration parameter. This portability is essential for the deployment migration path described in Section 8.

5.4 Data Sovereignty and Security Architecture

Genomic data processed by the platform is subject to the Protection of Personal Information Act (POPIA, No. 4 of 2013),[36] which classifies biometric and health information as special personal information and imposes heightened obligations on responsible parties handling such data. The NHREC Ethics in Health Research guidelines impose additional requirements on research involving human genomic data, including institutional oversight, informed consent documentation, and data governance protocols.[37]

The platform's data architecture is designed for POPIA compliance from inception. For iLifu HPC deployment, data never leaves UCT's institutional network: uploads pass directly to CBIO's secured storage, analysis executes within the institutional compute environment, and outputs are retained under CBIO's data governance policies. Cloud deployment options are architecturally configured to route clinical and patient-linked genomic data exclusively to South African data residency zones (currently AWS af-south-1 Cape Town region), satisfying the POPIA requirement for domestic data residency of special personal information. Research datasets without patient linkage can, at the investigator's discretion, be processed in any region with appropriate access controls.

6. The scRNA-seq Explorer: Design, Implementation, and Biological Rationale

6.1 The Single-Cell Imperative in African Infectious Disease Research

Single-cell RNA sequencing has fundamentally transformed our understanding of the immune response to Mycobacterium tuberculosis infection. Studies published between 2018 and 2024 using scRNA-seq have characterised the heterogeneous macrophage populations at the site of mycobacterial infection, identified transcriptionally distinct T-cell subsets associated with TB protection versus progression, and revealed the transcriptional correlates of trained immunity in BCG-vaccinated individuals. For Professor Mulder's TB host-pathogen biology programme — which seeks to characterise the human cellular response to M. tuberculosis at molecular resolution — scRNA-seq is not a future method to be adopted but a current necessity whose absence from CBIO's analytical toolkit represents an active research limitation.

The same applies across IDM and CIDRI's research programmes. HIV immunology requires single-cell characterisation of CD4+ T-cell depletion dynamics and viral reservoir formation. Malaria research employs scRNA-seq to characterise red blood cell invasion and immune evasion at single-cell resolution. Cancer biology uses single-cell profiling of the tumour microenvironment to identify immune checkpoint targets and predict immunotherapy response. In each case, the method is established; the limitation is access to a usable analytical pipeline at CBIO. The scRNA-seq Explorer provides that access.

6.2 The Eight-Stage Analytical Pipeline

The scRNA-seq Explorer implements the best-practices workflow described in Luecken and Theis (2019)[38] as an eight-stage guided analysis sequence. Each stage presents a curated parameter interface with evidence-based defaults, contextual documentation explaining the biological rationale for each parameter choice, and real-time output preview where computationally feasible. The design principle is progressive disclosure: novice users encounter sensible defaults that produce biologically reasonable outputs; experienced users can access the full parameter space by expanding advanced configuration panels.

Stage	Process	Biological Rationale	Key Parameters	Output
1	Data Upload & Validation	Ensure data integrity before computation	H5AD, MTX+barcodes, CSV; format auto-detection	Validated AnnData; cell/gene count summary;

Stage	Process	Biological Rationale	Key Parameters	Output
				sparsity statistics
2	Quality Control	Remove empty droplets, dying cells, doublets	Min genes/cell; max mitochondrial %; min cells/gene	QC violin plots; filtered AnnData; filtering summary table
3	Normalisation & Scaling	Correct for sequencing depth variation	Library size normalisation; log1p; HVG selection (n=2000)	Normalised matrix; HVG plot; expression distribution
4	Dimensionality Reduction	Extract principal axes of expression variation	PCA components (n=50); variance explained threshold	PCA scatter; elbow plot; PC gene loadings table
5	Neighbourhood Graph	Encode cell-cell transcriptional similarity	n_neighbors (default 15); representation (X_pca)	Neighbourhood graph metrics; connectivity statistics
6	Clustering (Leiden)	Identify transcriptionally distinct cell populations	Resolution (default 1.0); n_iterations	Cluster assignments; cluster size distribution; modularity score
7	UMAP Visualisation	2D projection for visual cluster inspection	min_dist (0.3); spread (1.0); colour by gene/cluster/metadata	Interactive Plotly UMAP; downloadable PNG/SVG/PDF
8	Cell Type Annotation & Report	Assign biological identities; document analysis	Marker gene sets; annotation confidence threshold; differential expression parameters	Annotated UMAP; dot plots; DE tables; pathway enrichment; automated PDF/HTML report

Table 3. The eight-stage scRNA-seq Explorer analytical pipeline, with biological rationale for each step.

6.3 Quality Control: Biology-First Design

QC filtering in scRNA-seq is a step where incorrect parameters produce systematically biased results — not obvious errors but subtle shifts in cell composition that propagate silently through all downstream analyses. The platform's QC stage is designed to prevent this through explicit biological framing. The three standard QC metrics — genes per cell (filtering empty droplets and multiplets), mitochondrial read fraction (filtering lysed or dying cells), and cells per gene (filtering unexpressed genes) — are presented with explanatory text that connects each metric to its biological meaning, not just its mathematical definition. Threshold sliders display real-time cell count projections, enabling researchers to see immediately how conservative or permissive filtering affects sample size.

The QC visualisation panel shows pre-filtering violin plots for all three metrics, allowing biological contextualisation of threshold choices: a tissue with high endogenous mitochondrial activity (cardiomyocytes, hepatocytes) will show naturally elevated mitochondrial fractions that should not be filtered at the standard 20% threshold appropriate for peripheral blood mononuclear cells. This biological context is documented in pop-up guidance panels that are specific to the tissue type entered by the user.

6.4 Clustering and Cell Type Annotation

The Leiden algorithm is implemented as the default clustering method, consistent with its demonstrated mathematical superiority over Louvain in guaranteeing well-connected communities.[39] The resolution parameter — which controls community granularity from a few broad clusters to many fine-grained populations — is presented with a live cluster count update, allowing researchers to calibrate granularity to their biological priors (the approximate number of expected cell types in their tissue).

Cell type annotation is the most biologically complex step in the pipeline and the one where automated assistance has the greatest potential to improve accuracy and reduce analyst burden. The current implementation supports three annotation approaches: manual annotation using marker gene dot plots with integrated CellMarker and PanglaoDB reference sets; supervised label transfer from reference datasets using Seurat-style anchor-based mapping; and a prototype integration with scANVI-based transfer learning for datasets where a labelled reference atlas exists. The AI integration roadmap in Section 9 describes the planned progression from these current approaches to foundation model-based annotation.

6.5 Automated Report Generation and Reproducibility

Every scRNA-seq analysis session generates two outputs beyond the visualisations: a structured PDF/HTML report and a JSON parameter configuration file. The report is formatted for direct inclusion in manuscript methods sections and grant applications — a deliberate design choice that makes reproducible documentation the path of least resistance rather than an additional task. The parameter configuration file captures every analytical choice made during the session, including software versions, parameter values, and the random seed used for stochastic steps (UMAP, clustering). Any analysis can be exactly reproduced from this file combined with the input data.

6.6 Live Deployment and Current Status

The scRNA-seq Explorer is currently deployed at scrna-analysis.streamlit.app, running on Streamlit Community Cloud as a proof-of-concept demonstrating the full eight-stage pipeline on user-uploaded AnnData objects. The deployment handles datasets up to approximately 100,000 cells within the Community Cloud resource constraints. The production deployment target — iLifu HPC via reverse proxy, described in Section 8 — removes this constraint, enabling analysis of the million-cell datasets characteristic of organ-level or atlas-scale studies.

7. The Variant Analysis Suite: African-Contextualised Genomic Interpretation

7.1 The African Variant Interpretation Problem

The systematic under-representation of African populations in global variant frequency databases is one of the most consequential limitations of current clinical genomics practice. The problem has been clearly described in the literature[40] but its practical implications for researchers and clinicians working with African patients are still not adequately addressed in standard variant interpretation tools. A variant with an overall gnomAD frequency of 0.01% may be benign population-specific variation in an African population with a sub-Saharan African-specific frequency of 2% — yet standard ACMG/AMP PM2 criteria would classify it as potentially pathogenic based on the overall frequency. The Variant Analysis Suite addresses this directly.

The suite implements African-contextualised variant interpretation as a first-class analytical feature, not an afterthought. Every variant classified by the ACMG-lite module is annotated with both the overall gnomAD allele frequency and the gnomAD African subpopulation frequency. Where these differ by more than one order of magnitude, the classification interface flags the discrepancy explicitly and recommends upgrading the PM2 criterion to benign evidence — providing the variant scientist with the information needed to make an appropriately contextualised interpretation. This is, to our knowledge, the first implementation of African-aware ACMG-lite classification in an open-source, accessible web interface.

7.2 Multi-Sample VCF Processing

The suite accepts multi-sample VCF files from any standard variant caller (GATK HaplotypeCaller, DeepVariant, Strelka2, bcftools call) as well as the direct output of CBIO's existing WGS/Exome pipeline. The upload interface performs format validation against VCF 4.2 specification and generates an immediate summary dashboard: sample count, total variant count, variant type distribution (SNVs, indels, MNPs, structural variants), quality metric distributions (QUAL, DP, GQ), and Ti/Tv ratio (a quality indicator for SNV calling accuracy). Anomalies in these metrics — low Ti/Tv ratio, excess low-quality calls — are flagged with contextual explanations to guide the researcher in deciding whether to proceed or to return to upstream pipeline QC.

7.3 Functional Annotation

Functional annotation is implemented through SnpEff[41] using the GRCh38.105 Ensembl annotation database, providing consequence prediction for coding, UTR, intronic, and intergenic variants across all annotated transcripts. The annotation pipeline reports the most severe consequence per variant using the VEP consequence

[41] Cingolani P et al. (2012). A program for annotating and predicting the effects of single nucleotide polymorphisms, SnpEff. Fly, 6(2), 80–92.

ranking hierarchy, with all alternative transcript consequences available through an expandable detail panel. Loss-of-function variants (stop-gained, frameshift, splice-site) are visually highlighted as the class requiring most careful pathogenicity consideration.

7.4 ACMG-lite Classification and Clinical Reporting

The ACMG-lite module applies computationally evaluable criteria from the Richards et al. (2015) framework[42], including PVS1 (predicted null variant in disease gene), PS1 (same amino acid change as established pathogenic), PM2 (absent/rare in population databases — African-contextualised as described above), PM4 (protein length change), PP3 (computational evidence of deleterious effect), BP4 (multiple algorithms predict benign), and BA1 (high allele frequency). Each criterion is displayed with its evidence basis, enabling variant scientists to audit the automated classification and apply expert judgement where automated evaluation is insufficient.

The clinical report generated for each variant session is structured to align with the output format recommended by the South African Society of Human Genetics (SASHG) and the NHLS genomics reporting standards — making the suite directly usable for clinical case reporting in South African diagnostic genomics contexts. This clinical alignment is a deliberate design choice: the suite is positioned not just as a research tool but as infrastructure for the emerging clinical genomics ecosystem in South Africa (HPCSA, 2021).

7.5 Population Genetics Module

For CBIO's African population genetics research programme, the suite includes a population genetics analysis module implementing: population-stratified allele frequency analysis across the major African ethnic groups represented in the African Genome Variation Project and H3Africa datasets; PCA-based population assignment using reference panel projection; Fst estimation for inter-population differentiation; and linkage disequilibrium analysis using scikit-allel. These tools are explicitly designed for African population genomics research — using African reference panels, African ethnic group labels, and African-specific population structure priors.

8. Deployment Architecture: South African Institutional Context and Data Sovereignty

8.1 The Institutional Deployment Problem

Deploying web-based bioinformatics applications in South African academic institutions involves constraints that are materially different from those facing commercial deployments or well-resourced Northern Hemisphere academic environments. UCT IT governance imposes strict controls on the exposure of internal systems to external networks. POPIA and NHREC guidelines constrain the routing of clinical genomic data through non-institutional infrastructure. The iLifu HPC cluster is designed and configured for batch job submission, not persistent web application hosting. Institutional IT procurement timelines and processes operate on scales incompatible with agile research software deployment.[43]

These are not surmountable through technical cleverness alone — they require institutional negotiation, policy engagement, and a deployment strategy that works within existing governance frameworks rather than circumventing them. This section presents the deployment analysis with that constraint explicitly in view.

8.2 Deployment Options: Comparative Analysis

Dimension	SSH Tunnel (Current)	Cloud PaaS (AWS af-south-1)	Reverse Proxy on iLifu	Managed K8s (EKS)
Public Accessibility	None — SSH required	Full — public HTTPS URL	Yes — via UCT DMZ proxy	Full — public HTTPS URL
Setup Complexity	Zero (existing)	Low — containerised deploy	Medium — UCT IT coordination	High — K8s manifests + ops
Data Sovereignty (POPIA)	Full compliance	Compliant (ZA region)	Full compliance	Compliant (ZA region)
HPC Pipeline Integration	Native — same filesystem	None — requires data transfer	Native — same SLURM queue	Via secure API only
Scalability	None	Auto-scaling (EC2/Lambda)	Limited (single node)	Excellent — horizontal

Dimension	SSH Tunnel (Current)	Cloud PaaS (AWS af-south-1)	Reverse Proxy on iLifu	Managed K8s (EKS)
Cost Model	UCT HPC allocation	Pay-per-use (~R2K–R8K/mo)	UCT HPC allocation	Reserved+spot (~R5K–R15K/mo)
Clinical Data Suitability	High	Conditional (ZA residency)	High	Conditional (ZA residency)
Recommended Phase	Deprecate (Phase 1)	Phase 1: Months 1–3	Phase 2: Months 4–8	Phase 3: Months 9+

Table 4. Comparative deployment analysis for CBIO's institutional context, incorporating POPIA data sovereignty requirements.

8.3 Phase 1 Recommendation: Cloud PaaS with South African Data Residency

For Phase 1 deployment, cloud PaaS on AWS af-south-1 (Cape Town) is recommended. This region provides South African data residency for POPIA compliance, sub-20ms network latency from UCT's campus network via SANReN, and full access to AWS managed services (RDS, S3, Lambda) that reduce operational overhead. The Streamlit application is containerised with Docker and deployed via AWS App Runner or ECS Fargate — a fully managed deployment that eliminates the need for UCT IT infrastructure coordination during the platform's initial demonstration phase. Research datasets without patient linkage can be processed immediately; patient-linked clinical data requires a formal data use agreement and dedicated VPC configuration, which should be initiated in parallel with Phase 1 development.

8.4 Phase 2 Recommendation: Reverse Proxy on iLifu HPC

Phase 2 deployment on iLifu via reverse proxy achieves the platform's long-term design goal: full integration with CBIO's HPC pipeline infrastructure, SLURM job submission, shared storage access, and on-premises data processing for all clinical data. The deployment requires: a DMZ-accessible server configured as an nginx reverse proxy (UCT IT coordination, 4–8 week timeline); a persistent Streamlit service managed by systemd on iLifu; SSL certificate via UCT's institutional certificate authority; and SLURM job status monitoring integration for the platform's progress reporting interface. The operational requirements are modest — this is a standard web application deployment behind institutional infrastructure — but the coordination overhead is non-trivial and should be initiated in Month 2 to be operational by Month 4.

8.5 Large Dataset Transfer Solutions

A persistent challenge for web-based bioinformatics platforms is the browser upload limitation that affects large genomics datasets. Single-cell datasets in H5AD format commonly range from 500 MB to 5 GB; whole-genome VCF files can exceed 10 GB; the raw FASTQ files that precede preprocessing can reach hundreds of GB. The platform addresses this through a dual-pathway data ingestion architecture: direct browser upload (suitable for pre-processed files under 2 GB, handled via Streamlit's file uploader with chunked transfer), and pre-staged upload to the user's iLifu workspace via SFTP or Globus, with the web interface referencing the staged file by path. For H3ABioNet training deployments where participants' data is pre-positioned by workshop organisers, batch dataset registration via a simple manifest file eliminates the upload bottleneck entirely.

9. Artificial Intelligence Integration: A Phased Roadmap

The integration of artificial intelligence into the platform is not a speculative aspiration — it is a near-term reality enabled by the maturation of biological foundation models and deep learning tools that are already available for production use. This section presents a structured roadmap for AI integration across three phases, grounded in the current state of the field and calibrated to CBIO's research requirements.

> *The question for CBIO is not whether AI will transform computational biology — it already has. The question is whether CBIO will deploy AI as a tool that amplifies South African research capacity, or whether African researchers will remain dependent on AI systems trained elsewhere, on non-African data, applying non-African biological priors.*

9.1 Phase 1 AI (Months 1–6): Automated Assistance Within Established Methods

Phase 1 AI integration focuses on automating the most time-consuming manual steps in the current analytical workflows using methods that are fully validated and immediately available.

9.1.1 Automated Cell Type Annotation

Manual cell type annotation — matching clusters to biological identities through marker gene inspection — is the most time-consuming and expertise-dependent step in scRNA-seq analysis. The current literature supports three automated annotation approaches that are ready for production deployment. Reference-based label transfer using Seurat's TransferData or the scANVI framework[44] projects new datasets into the embedding space of a labelled reference and assigns cell type labels based on nearest-neighbour similarity. For tissues well-represented in reference atlases (peripheral blood, bone marrow, liver), this approach achieves annotation accuracies above 90% against expert manual annotation. For TB-specific datasets, the Human Cell Atlas lung atlas and the single-cell TB host response datasets published by Cai et al. (2020) and Mbulaiteye et al. (2022) provide relevant reference material.

The scRNA-seq Explorer will integrate scANVI-based reference label transfer as an optional annotation step, with the user able to upload a custom reference dataset or select from CBIO-maintained reference atlases that will be curated specifically for the tissue types most common in IDM and CIDRI research (PBMC, lung, BAL, colon). Where confidence scores are below threshold, the interface flags cells for manual review rather than forcing an automated assignment.

9.1.2 AI-Assisted Variant Interpretation

For the Variant Analysis Suite, Phase 1 AI integration involves deploying computational pathogenicity prediction models beyond the rule-based ACMG-lite framework. CADD (Combined Annotation-Dependent Depletion), REVEL, and AlphaMissense — models that integrate evolutionary conservation, functional genomic evidence, and protein structure predictions — provide continuous pathogenicity scores that complement the categorical ACMG criteria. AlphaMissense (Cheng et al., 2023), trained on AlphaFold protein structures and population frequency data, achieves state-of-the-art pathogenicity prediction for missense variants. The platform will integrate these scores as additional evidence dimensions displayed alongside ACMG-lite criteria, with an AI-assisted evidence summary that synthesises the computational evidence and flags variants where different predictors provide conflicting signals.

9.2 Phase 2 AI (Months 6–18): Foundation Models and Contextual Biological Reasoning

Phase 2 AI integration deploys single-cell foundation models for biological tasks that exceed the capability of reference-based methods. This phase is contingent on the maturation and public availability of the relevant model infrastructure — a timeline that is credible given the pace of development in this field as of 2026.

9.2.1 scGPT and Geneformer for Zero-Shot Cell Type Annotation

Single-cell foundation models — scGPT (Cui et al., 2024), Geneformer (Theodoris et al., 2023), and CellPLM — are large transformer models pre-trained on tens to hundreds of millions of cells from diverse tissues and species.[45] Unlike reference-based methods, these models do not require a labelled reference dataset for the target tissue: they encode biological knowledge from their pre-training data that enables generalisation to novel cell types and tissue contexts. For African TB research datasets that may contain novel immune cell states or previously uncharacterised macrophage populations, this generalisation capability is particularly valuable — reference-based methods trained on non-African or non-TB data may fail to annotate accurately, whereas foundation model embeddings capture deeper biological representations.

Phase 2 integration will deploy scGPT embeddings as an alternative to PCA-based dimensionality reduction and Leiden clustering, enabling foundation model-based cell type annotation as an alternative analysis path. The interface will present both the standard Scanpy-based and the foundation model-based analyses side by side, allowing researchers to compare annotations and identify discrepancies that may indicate biologically novel populations meriting further investigation.

9.2.2 LLM-Assisted Biological Interpretation

Large language models fine-tuned on biomedical literature — BioGPT, BioMedLM, and the recent PubMedBERT-based models — can provide contextual biological interpretation of analytical results.[46] In Phase 2, the platform will integrate an LLM-based interpretation assistant that takes the cluster differential expression results as input and generates a literature-grounded interpretation: identifying which of the upregulated genes are known markers of a particular cell state, summarising the published evidence for that cell state in the tissue context, and flagging genes of interest for further investigation based on their known roles in the researcher's disease area.

This is not a replacement for biological expertise — it is an accelerator for the literature review and hypothesis generation steps that currently require hours of manual PubMed searching. The interpretation assistant is explicitly framed as a starting point for researcher investigation, with citations provided for every claim and a confidence indicator calibrated to the density of supporting literature. The outputs are designed to be critiqued and refined by the researcher, not accepted uncritically.

9.3 Phase 3 AI (Months 18–36): Autonomous Analysis and Adaptive Interfaces

Phase 3 represents the long-term trajectory of the platform toward genuinely intelligent analysis assistance. The technical prerequisites for this phase are partially satisfied by existing technology and partially dependent on near-future development.

9.3.1 Adaptive Parameter Optimisation

QC and analysis parameters that currently require researcher judgement — mitochondrial threshold, clustering resolution, UMAP parameters — can, in principle, be set automatically by models trained to predict optimal parameters from dataset characteristics. scAutoQC-style models that predict per-dataset QC thresholds from cell count distributions and tissue metadata are already demonstrated in the literature. Phase 3 will integrate adaptive parameter suggestion: the platform observes the dataset's statistical properties and proposes parameter values with a confidence range, which the researcher can accept, modify, or override. Over time, parameter choices made by expert researchers using the platform contribute to a CBIO-maintained training dataset that improves the adaptive parameter model for the specific dataset types (primarily PBMC, lung, and BAL in the IDM/CIDRI context).

9.3.2 Multi-Modal Analysis Integration

The scverse ecosystem's Muon framework extends AnnData to multi-modal data, enabling joint analysis of CITE-seq (RNA + protein), ATAC-seq (chromatin accessibility), and spatial transcriptomics datasets. Phase 3 will extend the scRNA-seq Explorer to support multi-modal data, with AI-assisted integration of modalities using MOFA+ or WNN-based methods. For CBIO's TB research specifically, the combination of single-cell transcriptomics with chromatin accessibility data (scATAC-seq) provides access to the regulatory landscape of immune responses — enabling identification of transcription factor activity and gene regulatory network changes during mycobacterial infection that are not visible from RNA alone.

9.3.3 Federated Analysis for H3ABioNet

A federated analysis architecture — in which H3ABioNet nodes contribute to shared analyses without transferring raw data — represents the most transformative long-term application of AI to the platform's continental mandate. Federated learning algorithms that train shared models from locally computed gradients, without requiring access to the underlying data, can enable collaborative analyses across African institutions while satisfying the data sovereignty requirements of each country's regulatory environment. Phase 3 will prototype federated analysis for cell type annotation: a shared model for PBMC cell type annotation trained collaboratively across UCT, Makerere University, Addis Ababa University, and other H3ABioNet nodes, without any institution sharing patient data beyond model updates.

Phase	Timeline	AI Capability	Technical Readiness	CBIO Research Benefit
Phase 1	Months 1–6	scANVI reference transfer annotation; AlphaMissense variant scoring; CADD/REVEL integration	Available now — production-ready libraries	Reduces annotation time by ~60–70%; improves variant interpretation accuracy for African variants
Phase 2	Months 6–18	scGPT/Geneformer foundation model annotation; LLM-assisted biological interpretation	Largely available — integration engineering required	Enables annotation of novel cell types without reference; literature-contextualised result interpretation
Phase 3	Months 18–36	Adaptive parameter optimisation; multi-modal analysis; federated H3ABioNet learning	Partially available — some components require development	Self-improving platform; continental collaborative analysis without data transfer

Table 5. AI integration roadmap across three phases, with technical readiness assessment and projected CBIO research benefit.

10. Evaluation Framework: Measuring What Matters

A research paper that proposes a new platform without a framework for evaluating its impact is a technology demonstration, not a research contribution. This section establishes the evaluation framework by which the platform's real-world effect — on researcher independence, analysis quality, reproducibility, and CBIO's research output — will be measured and reported.

10.1 Researcher Independence Metrics

The primary intended outcome of the platform is an increase in the number of UCT Health Sciences researchers who can conduct independent omics analysis without requiring bioinformatics support. This outcome will be measured through: (a) a baseline survey at platform launch documenting the number of researchers in each IDM/CIDRI department who have conducted independent genomic analysis in the past 12 months; (b) a follow-up survey at 6 and 12 months post-deployment documenting the same; and (c) IDM/CIDRI bioinformatician support request volume tracked before and after deployment. A meaningful platform effect should produce an increase in independent analysis accompanied by a reduction in routine support requests — not a reduction in bioinformatician engagement overall, but a shift in the nature of that engagement toward more complex, research-design-level support.

10.2 Analysis Quality and Reproducibility Metrics

Analysis quality will be assessed through a formal concordance study: a set of benchmark datasets with known cell type compositions (from published atlases) and known variant pathogenicity classifications (from ClinVar expert panel entries) will be analysed by (a) expert CBIO bioinformaticians using command-line pipelines, and (b) non-expert Health Sciences researchers using the platform. The concordance between the two will quantify whether the platform's guided analysis produces results of equivalent quality to expert manual analysis. This concordance study will be designed for publication as a standalone methods paper.

Reproducibility will be measured through a parallel reproducibility audit: a random sample of analyses conducted through the platform will be re-executed using the JSON parameter configuration files generated automatically at the end of each session, and the results compared against the original outputs. Perfect concordance (within floating-point numerical precision) is the expected and required result — any deviation indicates a reproducibility defect in the parameter recording system that must be resolved.

10.3 CBIO Research Output Metrics

The platform's effect on CBIO's research output will be tracked through: the number of publications from IDM and CIDRI that cite the platform in their methods sections; the number of grant applications that include the platform as part of their computational infrastructure evidence; and the number of H3ABioNet training participants who report using the platform independently after returning to their home institutions (assessed through 6-month post-workshop follow-up surveys). These metrics will form part of the platform's contribution to NRF RIPP and H3Africa grant applications, where demonstrated research infrastructure impact is a condition of funding.

11. Implementation Roadmap

Phase	Timeline	Milestone	Deliverable	Success Criterion
1	Month 1	scRNA-seq Explorer v1.0 + Variant Suite on Cloud PaaS	Live deployment at dedicated CBIO URL; IDM/CIDRI access enabled	≥10 IDM/CIDRI researchers onboarded; concordance study dataset prepared
1	Month 2	IDM/CIDRI structured pilot	Pilot with 5 IDM and 5 CIDRI researchers; structured feedback collected	Feedback report with prioritised improvements; >7/10 usability rating
1	Month 3	iLifu reverse proxy configuration	DMZ proxy configured with UCT IT; SSL certificate issued; SLURM integration tested	Platform accessible at cbio.uct.ac.za from public internet
1	Month 3	Phase 1 AI: scANVI + AlphaMissense integration	Annotation confidence scores in scRNA-seq Explorer; AI variant scores in Variant Suite	Concordance >90% against expert annotation on benchmark datasets
2	Month 5	Full CBIO pipeline dashboard layer	Interactive dashboards for all 6 existing CBIO pipelines	All pipelines accessible via web interface; output comparison to CLI results shows concordance
2	Month 6	TB genomics & metagenomics modules	M. tuberculosis comparative genomics dashboard; 16S rRNA visual explorer	Validated against existing CBIO TB dataset analyses
2	Month 8	Viral recombination visualisation	RDP-integrated recombination dashboard; phylogeographic mapping	Validated by Assoc. Prof. Martin's programme; RDP output concordance confirmed
2	Month 8	Phase 2 AI: scGPT integration +	Foundation model annotation path	Benchmark annotation accuracy; user evaluation of interpretation quality

Phase	Timeline	Milestone	Deliverable	Success Criterion
		LLM interpreter	in scRNA-seq Explorer; biological interpretation assistant	
3	Month 10	H3ABioNet training deployment	Platform deployed at 3+ H3ABioNet partner institutions; training materials published	Training participants from ≥3 African countries use platform independently post-workshop
3	Month 11	Open-source release	GitHub repository; Docker Hub images; full documentation; open-source licence	Repository publicly accessible; Docker images functional on standard HPC environments
3	Month 12	Grant applications with platform evidence	NRF RIPP, H3Africa, NIH Fogarty, Wellcome applications submitted	Applications incorporate platform metrics, concordance study results, and training impact data

Table 6. Full implementation roadmap with milestones, deliverables, and measurable success criteria.

12. Resource Requirements and Funding Pathways

12.1 Budget

Category	Item	Year-One Estimate (ZAR)	Justification
Platform Development	Interactive dashboards for 6 existing CBIO pipelines	R 20,000 – R 28,000	Integration with existing Nextflow/Docker pipelines; guided UI for each modality
Platform Development	scRNA-seq Explorer: 8-stage pipeline + AI annotation	R 12,000 – R 18,000	Including Phase 1 AI integration (scANVI, AlphaMissense)
Platform Development	TB & pathogen modules; viral recombination visualisation	R 10,000 – R 15,000	Comparative genomics dashboards; RDP integration; metagenomics explorer
Infrastructure	Docker/Singularity containers; Nextflow config; iLifu integration	R 3,000 – R 8,000	Container publishing; SLURM integration; reverse proxy configuration
Workstation	High-performance development and analysis workstation	R 28,000 – R 42,000	GPU-capable for AI model integration; local development/testing
Cloud Compute	AWS af-south-1 Phase 1 deployment; model inference	R 4,000 – R 10,000 /yr	EC2 t3.medium–t3.large; S3 storage; Phase 1 compute costs
Researcher Stipend	Full-time development, integration, evaluation, publication	R 216,000 – R 300,000 /yr	MSc-level researcher; 12 months; Cape Town market rate
Training	H3ABioNet workshop integration; training materials; travel	R 6,000 – R 12,000	Workshop participation at CBIO and 1–2 partner institutions
Hosting & Maintenance	Platform hosting; pipeline updates;	R 8,000 – R 14,000 /yr	Server costs; dependency updates; user support

Category	Item	Year-One Estimate (ZAR)	Justification
	ongoing improvements		

Table 7. Itemised year-one budget. All figures in ZAR, 2026. Ongoing annual costs (compute, maintenance, stipend) apply beyond year one.

Budget Line	Year-One Estimate (ZAR)
Platform development (all components)	R 42,000 – R 61,000
Infrastructure & containerisation	R 3,000 – R 8,000
High-performance development workstation	R 28,000 – R 42,000
Cloud compute (Phase 1 deployment)	R 4,000 – R 10,000
Researcher stipend (12 months)	R 216,000 – R 300,000
Training, hosting & maintenance	R 14,000 – R 26,000
Year-One Total	R 307,000 – R 447,000
Multi-Year Total (including ongoing)	R 480,000 – R 680,000+

12.2 Funding Pathways

Funder	Programme	Alignment Rationale	Competitive Position
NRF	Research Infrastructure Programme (RIPP); SARIR; Thuthuka	Platform as national bioinformatics infrastructure serving NICIS strategy; directly fundable under RIPP Category B (domain-specific research platforms) — see NRF 2023	Strong — platform fills documented gap in SA research infrastructure; aligned with DSI White Paper priorities
H3Africa / H3ABioNet	Pan-African genomics training and analysis	Platform extends CBIO H3ABioNet node mandate; directly serves H3Africa data analysis requirements; training impact measurable across network (Mulder et al., 2018)	Very strong — CBIO as primary SA H3ABioNet node; platform addresses documented H3ABioNet capacity gap

Funder	Programme	Alignment Rationale	Competitive Position
NIH Fogarty	African health research capacity building (D43, D71)	Platform as training infrastructure evidence in capacity-building applications; scRNA-seq and variant analysis serve NIH infectious disease priorities (TB, HIV)	Strong — demonstrated platform infrastructure combined with CBIO's research track record
Wellcome Trust	African genomics; infectious disease; capacity building	TB and population genomics modules serve Wellcome's Africa programme; CBIO's IDM partnership aligns with Wellcome's African institutional investment strategy	Strong — Wellcome's existing IDM funding creates natural platform for platform investment
UCT Research Development	Internal seeding; innovation grants	Phase 1 prototype funding; proof-of-concept demonstration for external grant applications	Moderate — competitive with other UCT internal applications; strongest case if CBIO formally endorses
IDM/CIDRI	Institutional bioinformatics support budget	Platform reduces routine support burden; frees bioinformatician time for complex research; directly serves IDM/CIDRI research programmes	Strong — direct institutional benefit; IDM/CIDRI endorsement strengthens all external applications

Table 8. Funding pathways with competitive positioning analysis.

13. Conclusions: The Platform as Epistemic Infrastructure

13.1 What This Paper Establishes

This paper has argued — and demonstrated — that the computational access barrier in African genomics is not a usability problem amenable to better documentation or a friendlier interface. It is a structural inequity in the distribution of analytical capacity that determines who can extract knowledge from genomic data, and therefore whose scientific questions get asked, whose populations get studied in depth, and whose researchers build the expertise to lead the next generation of biomedical science. The platform described in this paper is a response to that structural condition in its most tractable, immediately addressable form: at CBIO, UCT, where excellent analytical infrastructure exists but reaches only a fraction of the researchers who need it.

The contributions of this paper are five. First, an empirically grounded analysis of the bioinformatics capacity gap in South African and African research institutions, situating the platform's rationale in the peer-reviewed literature and in national science policy documents. Second, the design and production deployment of two novel analytical systems — the scRNA-seq Explorer and the Variant Analysis Suite — that directly address documented gaps in CBIO's analytical toolkit and the broader South African research infrastructure. Third, an African-contextualised variant interpretation framework that explicitly addresses the systematic bias introduced by European-dominant population frequency databases — a research contribution independent of the platform itself. Fourth, a rigorous deployment architecture analysis that incorporates South African data sovereignty law (POPIA), the NICIS cyberinfrastructure strategy, and UCT's institutional computing constraints. Fifth, a detailed AI integration roadmap that positions the platform for a trajectory from current production capability to foundation model-based biological reasoning over a defined 36-month horizon.

13.2 The Larger Stakes

Behind the platform architecture and the technology stack, there is a larger argument about the nature of African scientific participation in the genomics era. The human genome — including the genome of every South African patient who consents to genomic research — contains biological information of extraordinary value. The institutions with the capacity to interpret that information determine the knowledge that is produced from it. If African institutions cannot independently analyse African genomic data — if they must route their data through external platforms, external pipelines, and external interpretive frameworks — then the knowledge produced from African biology will continue to reflect the questions, priorities, and analytical assumptions of researchers elsewhere.

The platform described in this paper is a contribution to reversing that dynamic. It is modest in scale — a web application, a set of analytical modules, a deployment architecture. But its significance is in what it makes possible: a UCT Health Sciences researcher who could not previously analyse their own scRNA-seq data can now do so, independently, reproducibly, with results that are as methodologically rigorous as those produced by a command-line expert. That researcher's scientific questions, their interpretations, their hypotheses — all become possible that were previously foreclosed. Multiplied across 14 clinical departments, across H3ABioNet's continental network, across the annual cohort of CBIO-trained bioinformaticians returning to institutions across Africa — the platform's contribution is to the epistemic sovereignty of African science.

> *CBIO is already doing the right work. The pipelines exist. The research programmes are world-class. The continental network is real. What has been missing is the interface that makes all of it accessible to every researcher who needs it — not just those fluent in the command line. This platform provides that interface. What follows, for CBIO and for African genomics, is what those researchers will discover when they can finally ask their own questions with their own data.*

13.3 Future Directions

Beyond the 36-month AI integration roadmap, several longer-horizon research directions emerge from this work. Multi-institutional federated analysis — enabling collaborative genomic studies across H3ABioNet nodes without data transfer — requires protocol development, ethical framework negotiation across multiple national jurisdictions, and technical architecture that is only partially specified here. African-specific foundation model development — training single-cell and variant models on African genomic data to correct the systematic representation bias in currently available models — requires the data aggregation, curation, and governance infrastructure that the H3Africa consortium is positioned to provide, with CBIO's platform as a natural endpoint for model deployment. Clinical genomics integration — extending the Variant Analysis Suite to support NHLS genomic reporting standards and HPCSA ethical guidelines for clinical genomic practice — positions the platform as infrastructure for South Africa's emerging clinical genomics ecosystem.

Each of these directions is a research programme in its own right. Each depends on the foundation established by the platform described in this paper. The work is not complete. It is begun.

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Appendix A: Live Platform Resources

Resource	URL	Status	Capabilities
scRNA-seq Explorer	scrna-analysis.streamlit.app	Production-deployed	8-stage guided pipeline: upload → QC → normalisation → PCA → clustering → UMAP → cell type annotation → DE analysis + automated PDF/HTML report
Variant Analysis Suite	simon-variants.streamlit.app	Production-deployed	Multi-sample VCF: SnpEff annotation, ClinVar integration, African-contextualised ACMG-lite, HTML clinical report, population genetics module
GitHub Repository	github.com/Simon-Mufara	Public	Source code, Docker configurations, Nextflow pipeline scripts, QC container (simonmufara/qc_and_trim, 60+ pulls)
Docker Hub	hub.docker.com/u/simonmufara	Public	Production QC container (qc_and_trim); Singularity-compatible; used in CBIO pipeline integration
ORCID Record	orcid.org/0000-0002-4430-2380	Active	Verifiable research output record; authorship verification

Appendix B: Author Profile

Simon Mufara is an MSc Computational Health Informatics candidate at UCT's Division of Computational Biology — an internal UCT researcher who brings both production software engineering capability and deep familiarity with CBIO's research environment, pipeline infrastructure, and scientific culture.

The two production-deployed platforms documented in this paper — scrna-analysis.streamlit.app and simon-variants.streamlit.app — demonstrate not the intention to build this capability but the fact of its existence. Both platforms are functional, accessible, and already usable by CBIO researchers. The Docker Hub-published QC container (simonmufara/qc_and_trim, 60+ pulls) demonstrates production-ready containerisation aligned with CBIO's HPC infrastructure. This paper is therefore a report on work already accomplished and a proposal for its systematic extension — not a prospectus for future activity.

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